Hyojun Jeong

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Education

Carnegie Mellon University — Pittsburgh, PA
M.S. in Mechanical Engineering — Aug 2026 - May 2028
B.S. in Mechanical Engineering — Aug 2018 - May 2022

Experience

CMU Robomechanics Lab — Pittsburgh, PA
Graduate Research Student — Aug 2026 - Present

- Designing a tendon-driven wing-retraction joint for a bio-inspired morphing-wing glider, comparing tendon and linkage concepts across 3 rapid prototypes
- Selecting actuators and integrating them with onboard electronics and embedded control for the morphing wing within a strict mass budget
- Optimizing landing trajectories and morphing-wing control for the glider

General Motors — Warren, MI
Virtual Design Development and Validation Engineer — May 2022 - Aug 2026

- Led rapid development of a universal steering wheel armature used on many production vehicles, running structural and durability FEA in LS-DYNA and OptiStruct
- Ran FMVSS 201 head impact load cases for occupant safety on other programs
- Visited supplier test sites as CAE point of contact for 2 high-priority programs, reviewing physical test setups and correlating FEA simulations with test results
- Built a PhysicsAI tool to predict steering system modal (vibration) performance, cutting HPC demand and manual effort by 25% via sensitivity analysis of 42 variables

CMU Jayan Lab — Pittsburgh, PA
Undergraduate Research Assistant — Jan 2022 - May 2022

- Designed a SolidWorks fixture to hold a spectrophotometer probe for a chemical-handling robot, fabricating it with 3D printing and laser cutting
- Integrated the probe fixture and spectrophotometer with the robot, debugging hardware and software issues during integration

Projects

CMU Moon Miners (NASA Lunabotics) — Pittsburgh, PA
Structures Team — Nov 2026 - Present

- Contributing to the structural design of an autonomous lunar excavation rover for the 2027 NASA Lunabotics Challenge, through preliminary and critical design reviews
- Running FEA on structural components to verify strength and stiffness under excavation loads, vibration, and harsh, dusty regolith terrain

MoonRanger — Pittsburgh, PA
Chassis Team — Jan 2021 - Dec 2021

- Designed a 24 lb carbon fiber chassis for an autonomous lunar rover, balancing mass, stiffness, and thermal performance for an extreme environment
- Packaged electronics and sensors into a high-density, suitcase-sized layout, iterating CAD assemblies to resolve fit and clearance
- Validated the design with 20+ structural and thermal simulations in ANSYS and SolidWorks covering launch vibration, lunar terrain loads, and thermal extremes

CAD: SolidWorks, Siemens NX, Fusion 360, GD&T drawings
CAE / FEA: ANSYS (structural, thermal), LS-DYNA, OptiStruct, HyperMesh, ANSA, nCode
Fabrication: CNC, mill, lathe, bandsaw, drill press, 3D printing, laser cutting
Programming and Embedded: Python, C++, MATLAB, Arduino, Raspberry Pi
AI/ML and Robotics: PyTorch, Isaac Lab, MuJoCo